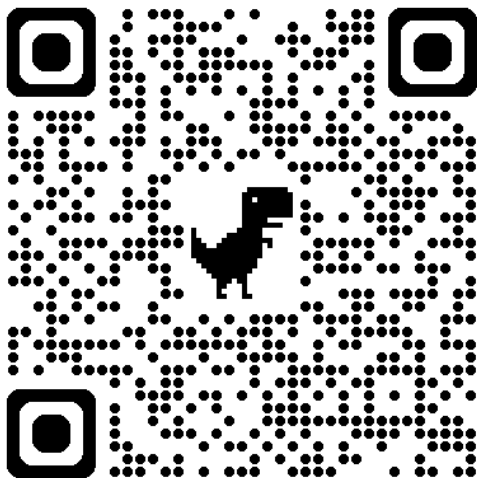


Vector

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1. Recap Problems

2. Vector

3. Practice Problems

Recap Problems

Vector

What is Vector?

In mathematics, a vector is an object that has both magnitude and direction. But in programming, a vector is a dynamic array that can resize itself automatically when elements are added or removed. It provides random access to its elements and is implemented as a contiguous block of memory.

You can consider vector as a flexible array that can grow or shrink in size as needed. It is a part of the C++ Standard Template Library (STL) and is widely used for storing and manipulating collections of data.

Declaration of vector

```
1      // declaration:
2      vector<TYPE> name;
3      // declaration with size:
4      vector<TYPE> name(SIZE);
5      // declaration a multidimensional vector:
6      vector<TYPE> name[SIZE];
```

Note that if you declare a vector w/ `vector<TYPE> name`, it will be initially empty.

Usage of vector

```
1      // access elements:
2      name[index];
3      // add elements:
4      name.push_back(value);
5      name.emplace_back(value);
6      // remove elements:
7      name.pop_back();
```

Usage of vector

```
1      // access elements:
2      name[index];
3      // access it's size:
4      name.size();
5      // remove all elements:
6      name.clear();
```

Example

```
1      vector<int> v;  
2      v.push_back(1);  
3      v.push_back(2);  
4      v.push_back(3);  
5      for(int i=0;i<v.size();++i){  
6          cout<<v[i]<<" ";  
7      }
```

Practice Problems
